

LECTURE NOTES: LIMITS AND CONTINUITY

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ABSTRACT. A brief introductory exposition on limits, continuity, and related theorems.

1. EXAMPLES OF LIMITS

We begin with an intuitive definition of limits of real-valued functions.

Definition 1.1. For a real-valued function $f : D \rightarrow \mathbb{R}$ defined on some subset $D \subseteq \mathbb{R}$, we say that *the limit of $f(x)$ as x approaches a is L* if the value of $f(x)$ gets arbitrarily close to L as x gets closer and closer to a , without necessarily reaching a .

That is, the existence of a limit depends only on the behavior of the function near the point a , regardless of whether $f(a)$ is defined. Since we can approach a from both sides (either right or left) on the real number line, for the overall limit to exist, the *right-hand limit* $\lim_{x \rightarrow a^+} f(x)$ and *left-hand limit* $\lim_{x \rightarrow a^-} f(x)$ must both exist and equal the same value L .

Here we look at some “nontrivial” examples (or counterexamples) of limits of functions.

Example 1.2. Let $f(x) = \sin \frac{1}{x}$. If we ever try to estimate the limit $\lim_{x \rightarrow 0} f(x)$ by either graphing or calculating $f(x)$ for very small values of x , we quickly run into an issue; the function seems to “fluctuate” across many different values as we take $x = 0.01, 0.00001, 0.0000001$, and so on.

In fact, the function indeed oscillates *infinitely often* on any open interval containing the point 0. In particular, if we let

$$x_1 = \frac{1}{\frac{1}{2}\pi + 2n\pi}, \quad x_2 = \frac{1}{\frac{3}{2}\pi + 2m\pi},$$

then any open interval around 0 contains the points x_1 and x_2 for sufficiently large n and m , while $f(x_1) = 1$ and $f(x_2) = -1$, meaning that the function oscillates between these two values (which are indeed upper and lower bounds of f).